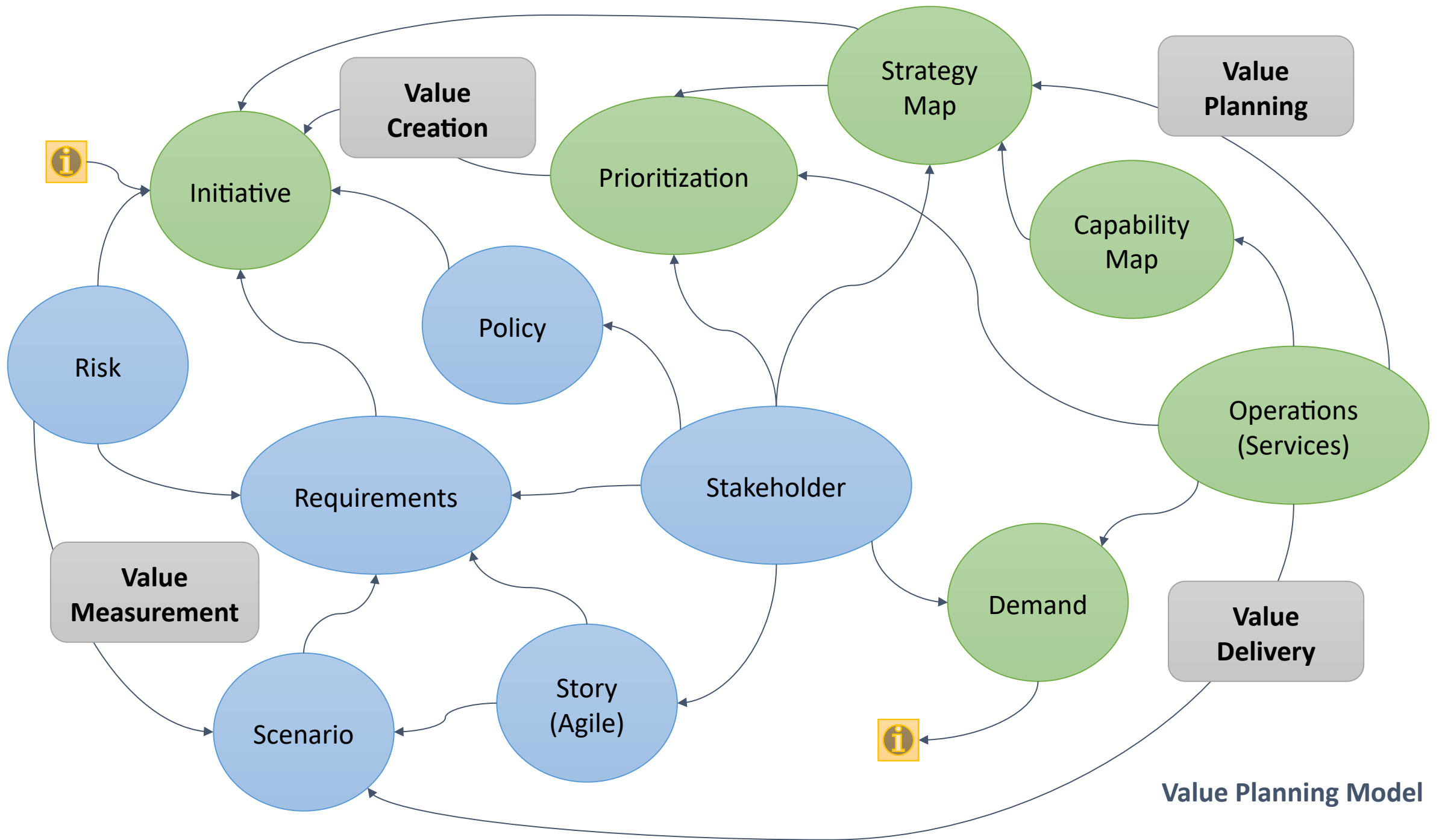


Future
State

Model Driven Services Management

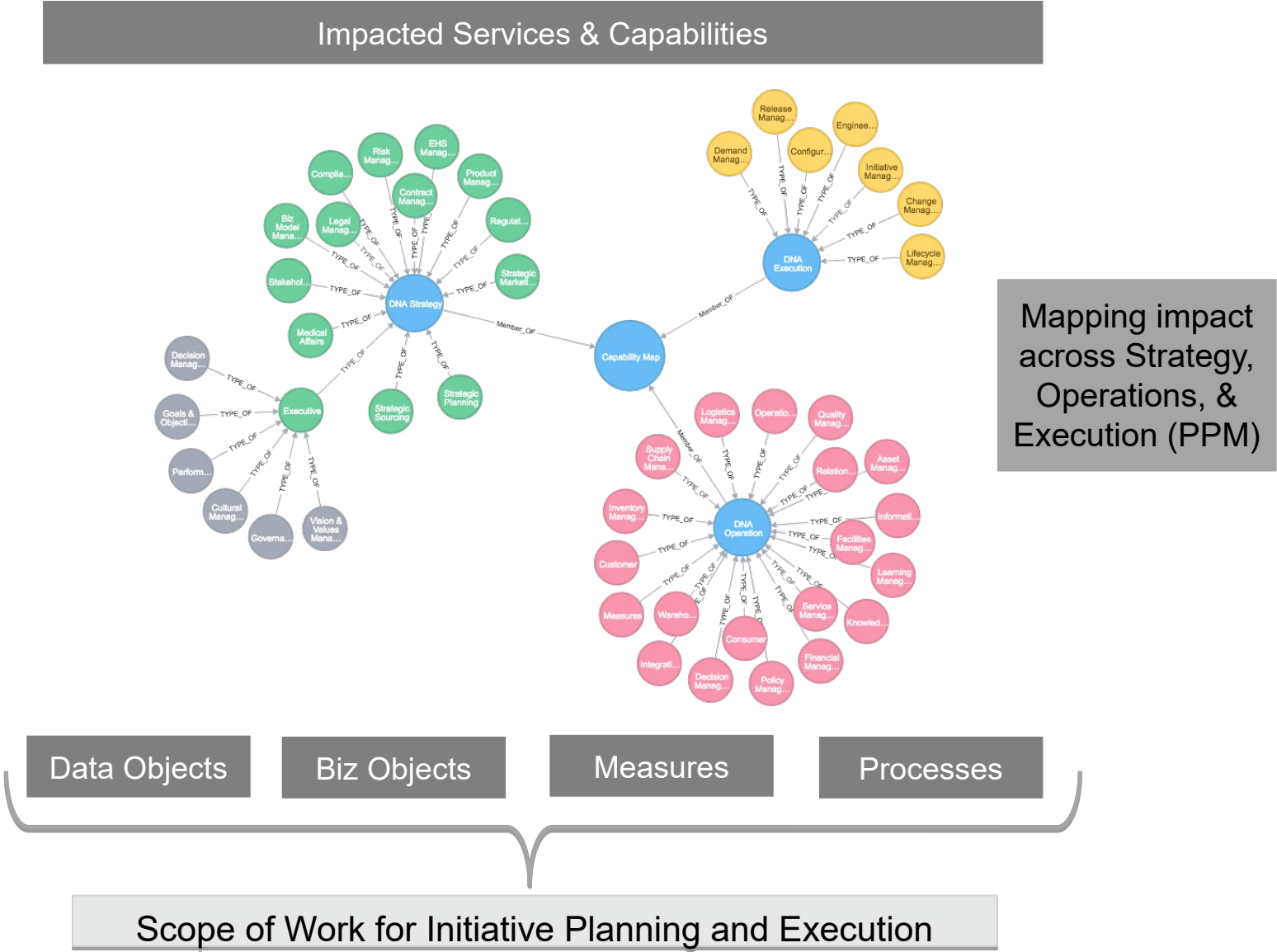
Extending Integration & DAG's Operational Model using KGs



Value Planning Model

Service Planning Model (Future State)

Value Chain	Inspect Transmission Tower
Capabilities	Tower Inspection (L3) OPS Maintenance (L2) OPS Support (L1)
Data Objects	Asset Ref Model Item & Parts Assembly Issue / Defect Stakeholders
Business Objects	Policy (numerous...) Maintenance Manuals Service / Task Tickets Schedule & Projects
Measures	Commitments Scheduling Customer Impact Upsell Opportunities
Models	Predictive Failure Repair / Replace Capacity Optimization Resource Utilization
Processes & Rules	Schedule Main. slot Schedule Maint. resources Access Stakeholder Info Update Service info



Governance Framework for Service Delivery

Enterprise

Structures

- Ontologies
 - Concepts, Content, Change
 - Context, Relationships
- Taxonomies
 - Enterprise Glossary

Domain-Specific

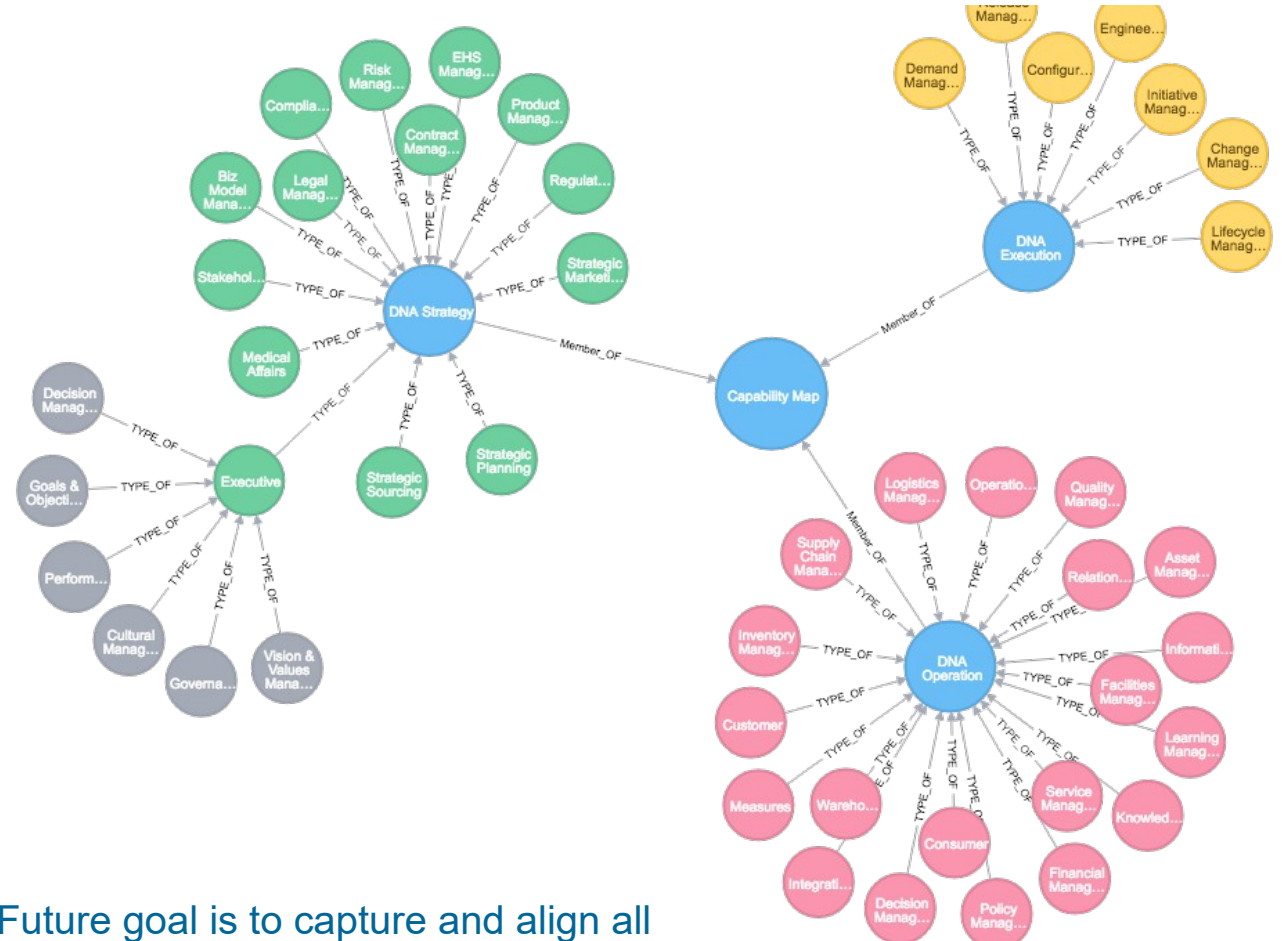
Data

- Scenarios
- Context
 - Relationships
 - Properties
 - Lifecycle (change)
 - Lineage
- Semantics
- Local Glossaries

Shared Services

Services

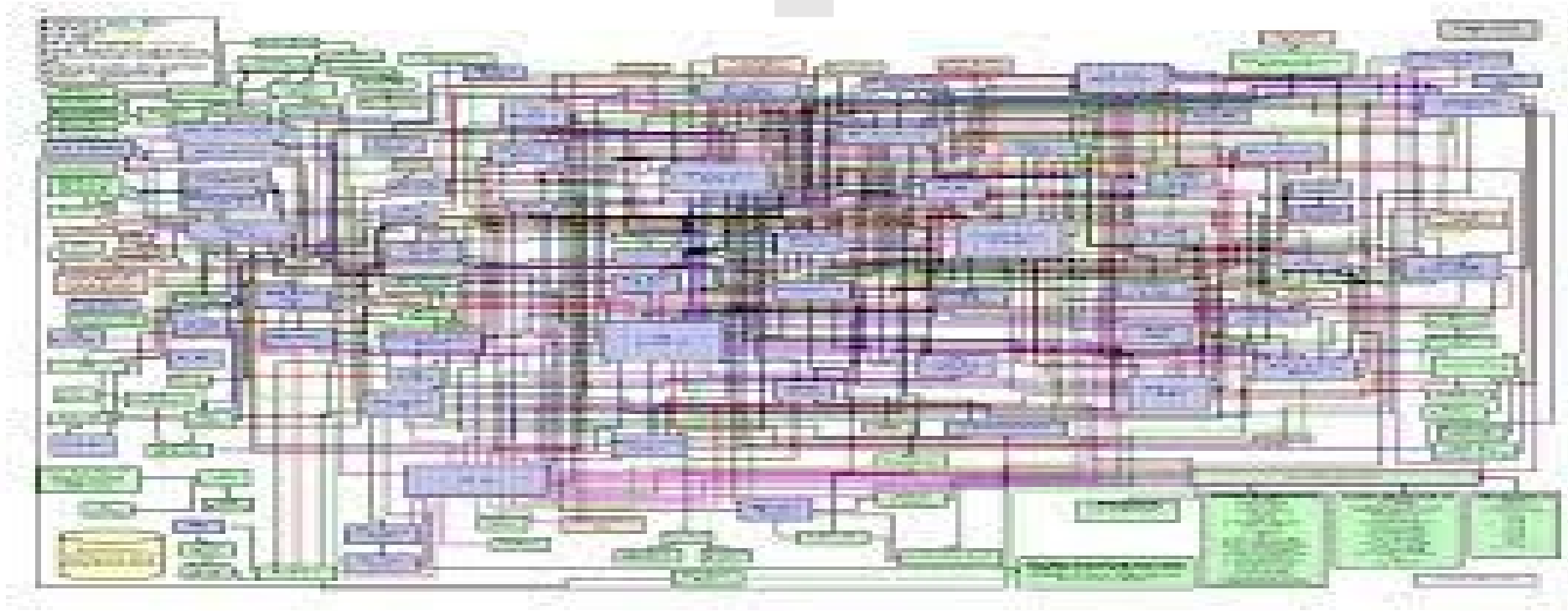
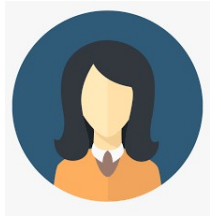
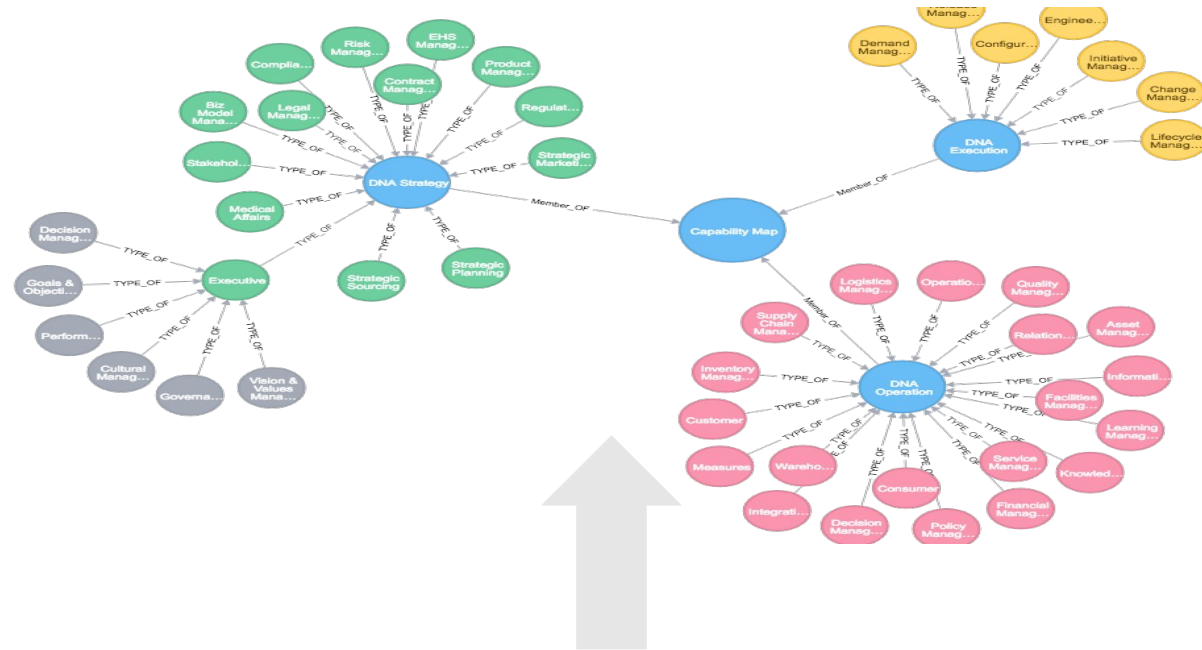
- Integration & Sourcing
- Search (Boolean, NLP)
- Analytics (Quality, Insights)
- Visualization (Navigation)
- Integrity & Security
- Publishing (APIs, Data Sets)
- Auditing & Utilization



Organic Solution

(Simplifying Search, Discovery, & Insights)

Knowledge Graphs
provide a scaffolding that
sits atop existing systems.

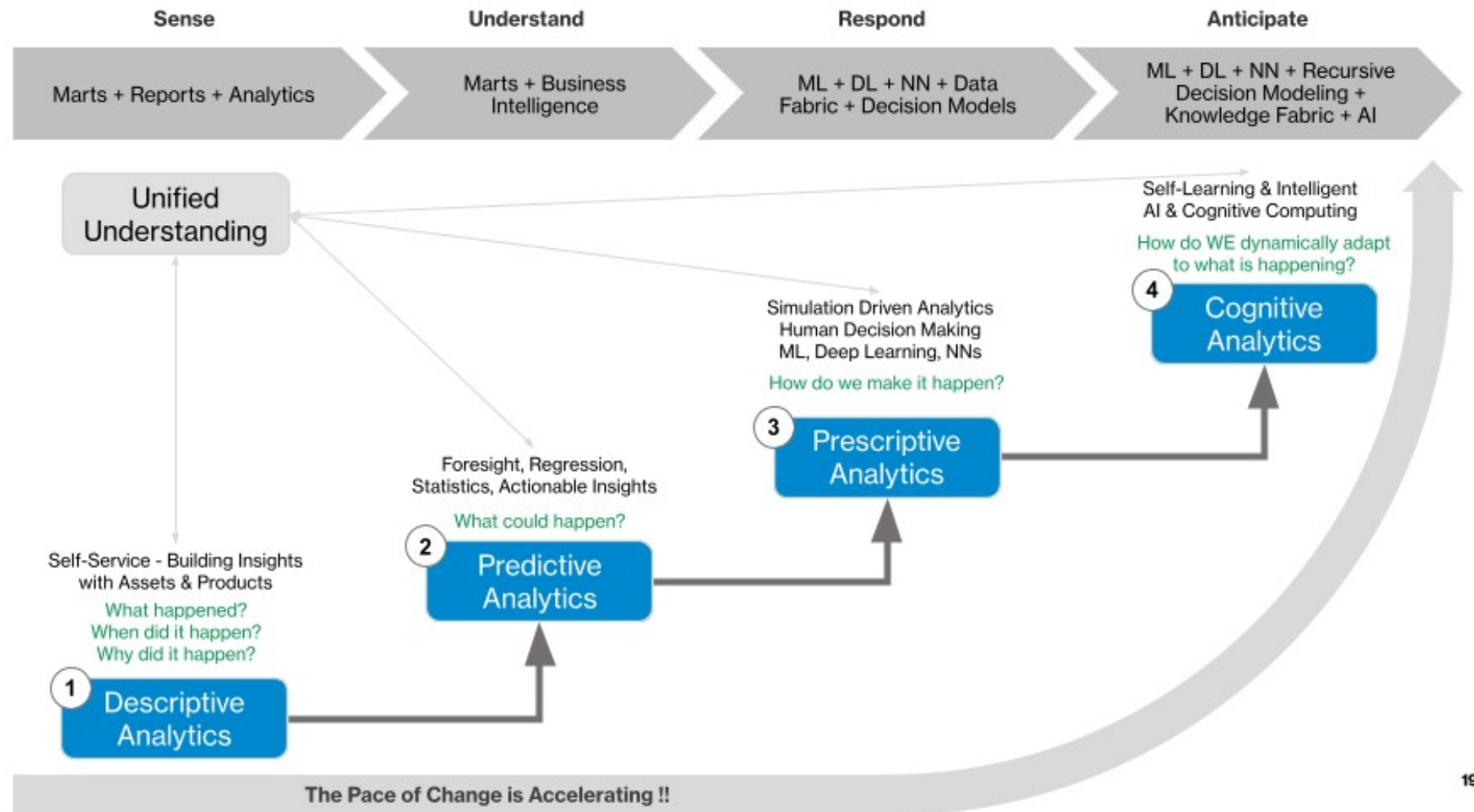


Knowledge-Driven Governance

now we can “link” changes to any artifact (*doc, code, service, outcome, feature, use case, design specification, SOP*) to any other artifact within IT, across our data, across the business.

Understanding the impact of change is the first step towards managing and responding to change at the speed of business.

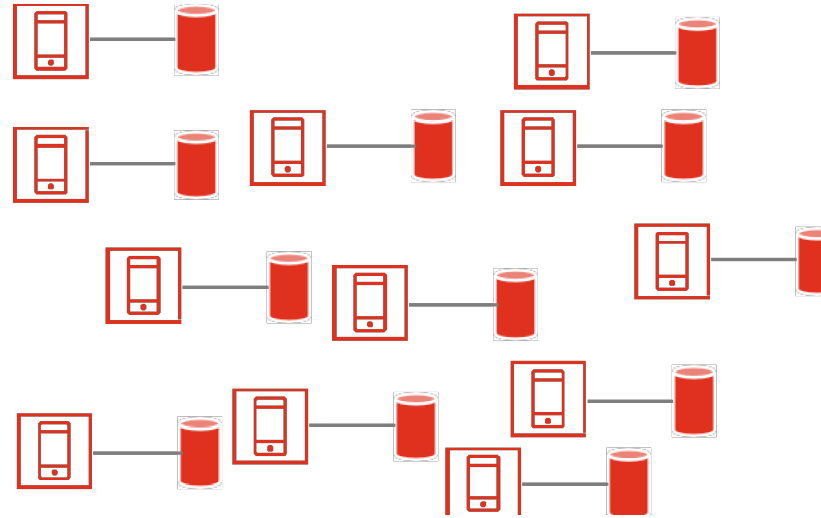
Roadmap for Accelerating Decision Making & Business Agility



Key Benefits of KG adoption

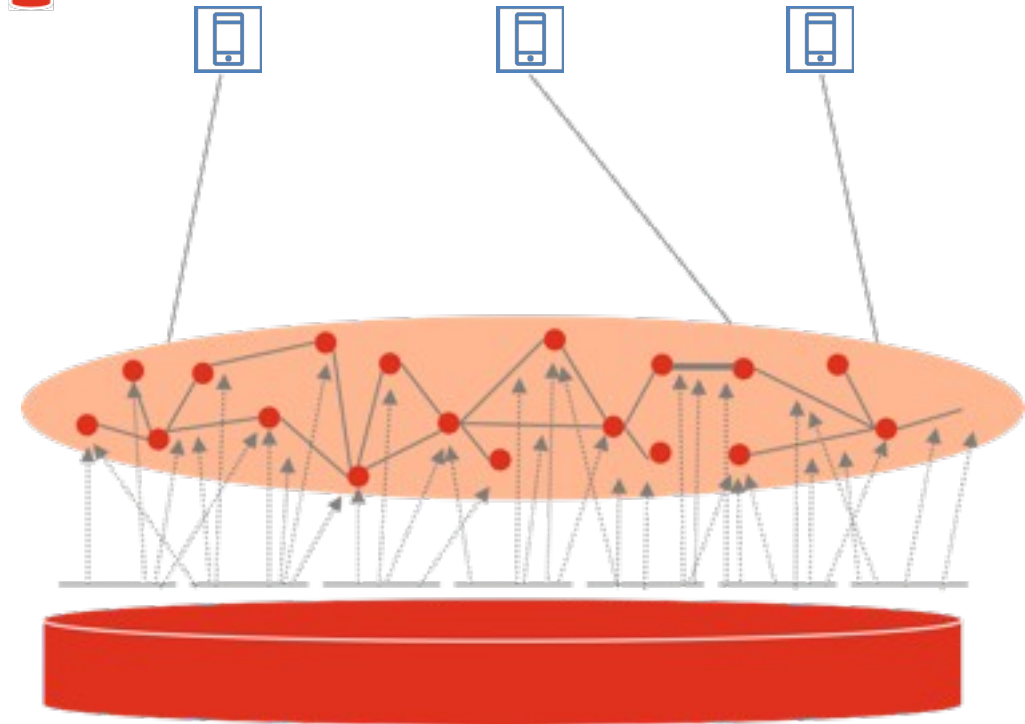
- Combines Conceptual & Logical Views of data
- Preserves enterprise and community perspectives of “truth”
- Shifts spend on fixing existing problems so we can solve new ones
- Increases adaptability and resiliency for consumers of data & information
- Provides unlimited scale to manage volume and complexity
- Integrates the lifecycle and operational management (systems, products, services, data, analytics)
- Provides a foundation for designing and managing Digital Twins while improving AI and leveraging ML for knowledge capture & consumption
- Accelerates innovation and cycle time delivery for new products & services

Shift from app-
centric to
data-centric value
delivery



Services built around
application data using fixed
perspectives & semantics

Services built
around business
need & semantics



Advantages of embracing KGs

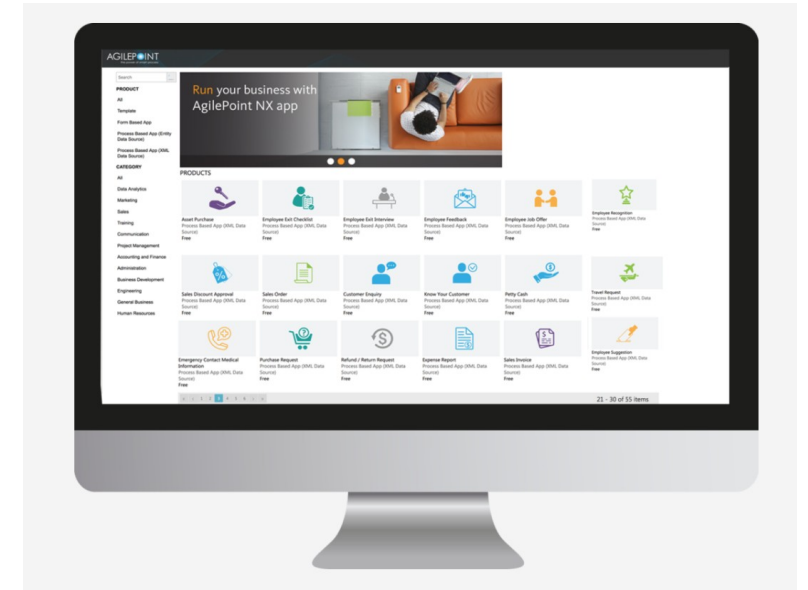
The 4Ds

- **De-Silo**
 - Integrate all the relevant sources in a declarative fashion that enables reuse, cross-enterprise scalability, and continuous refinement.
- **Disambiguate**
 - Disambiguate: Triangulate using set theory and linguistic description logic in addition to statistical methods, enabling precise entity resolution.
- **Detect**
 - Detect: Uncover weaker signals by articulating the most relevant and distant relationships between entities, via richer contextualization.
- **Discover**
 - Discover: Radically expand the ability to discover insights, moving beyond keywords to concepts.

Towards a Future State of Self Sufficiency

Self sufficient process automation built atop a set of published libraries:

- Process (Activity) Components
- Data Services (1)
- Integration Services (2)
- Event Management Services (3)
- Deployment Services
- Reporting & Analytic Services



- (1) Data management, data quality, data transforms, semantic matching
- (2) Data and Process integration with loose coupling, highly available, with ability to deliver QoS to address scale & quality
- (3) Manually triggered workflow, timed events, and automatically triggered activities